

# Estimating Beta from CAPM and Volatility

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## Keywords

beta, volatility, linear regression

## 1 Introduction

From the Capital Asset Pricing Model (CAPM) of Sharpe (1964), Lintner (1965), and others, we have if  $R_{i,t}$  is the return on the security  $i$ ,  $R_{f,t}$  is the return on the riskless security, and  $R_{m,t}$  is the return on the market portfolio, then

$$\mathbb{E}[R_{i,t}] = R_{f,t} + \beta_i (\mathbb{E}[R_{m,t}] - R_{f,t}).$$

where

$$\beta_i = \frac{\text{Cov}(R_{i,t}, R_{m,t})}{\text{Var}(R_{m,t})}.$$

A stock's market beta can be estimated using the following CAPM or one-factor market model regression:

$$r_{i,t} = \alpha_i + \beta_i \cdot \text{MKT}_t + \varepsilon_{i,t},$$

where  $r_{i,t}$  is the excess return of stock  $i$  during period  $t$ ,  $\text{MKT}_t$  is the excess return of the market factor or portfolio during period  $t$ , and  $\varepsilon_{i,t}$  is the regression residual. This estimation involves performing a linear regression, where the dependent variable is the excess return of stock  $i$ , and the independent variable is the excess return of the market.

Given the market model estimation, we are able to decompose the total volatility of an individual stock into the systematic volatility and the idiosyncratic volatility as follows:

$$\sqrt{\text{var}(R_{i,t})} = \beta_{i,t} \sqrt{\text{var}(R_{m,t})} + \sqrt{\text{var}(\varepsilon_{i,t})}.$$

That is, the total volatility of stock  $i$ ,

$$\text{TVOL}_{i,t} = \sqrt{\text{var}(R_{i,t})},$$

is a sum of the systematic volatility,

$$\text{SVOL}_{i,t} = \beta_{i,t} \sqrt{\text{var}(R_{m,t})},$$

and the idiosyncratic volatility,

$$\text{IVOL}_{i,t} = \sqrt{\text{var}(\varepsilon_{i,t})}.$$

Systematic risk refers to the broader market risk that will affect all investments, while idiosyncratic volatility refers to risk specific to a firm.

## 2 Implementation

For this project, data from the CRSP dataset was used, specifically from 1996 to 2023. The CRSP dataset provides detailed security-level identifiers, price information, trading activity, and return measures used in empirical asset pricing analysis. Each security is uniquely identified by a permanent numerical identifier (PERMNO) and is linked at the firm level through a permanent company identifier (PERMCO). Share codes (SHRCD) classify securities by share type, while industry affiliation is captured through Standard Industrial Classification (SIC) codes. Firm names and trading symbols are provided via company names (COMNAM) and ticker symbols (TICKER), and securities are further referenced using CUSIP identifiers. Daily market information includes transaction prices (PRC), bid and ask quotes (BID, ASK), and daily low and high quote bounds (BIDLO, ASKHI), which together characterize prevailing market valuations and liquidity conditions. Trading activity is measured by daily share volume (VOL) and shares outstanding (SHROUT), allowing for the construction of market capitalization and liquidity proxies. Return measures include raw security returns (RET), returns excluding distributions (RETX), and the value-weighted market return (vwret), which serves as a benchmark for market-wide performance in asset pricing tests.

The first step for the project was to clean the data. A PreProcessor class was created which could remove non-numeric values from a given column, map SIC codes to specific industries, extract T-Bill data as a proxy for risk-free interest rates, and create a random sample of 10 firms per industry for each year between 1996-2023.

The next step was to build a variety of features. A FeatureEngineer class was built that would be able to take the sample built by PreProcessor and build features related to excess returns, a beta based on rolling windows of various lengths, and breakdown the total volatility into systematic and idiosyncratic volatilities.

The last two steps primarily aided in the analysis of FeatureEngineer. The Analysis class was able to calculate descriptive statistics on the betas, volatilities, as well as quintile portfolios based on various sorts applied to the dataset. The Visualization class created the visualizations used in this report. The repository for the implementation of this project can be found [here](#).

### 3 Results and Analysis



Figure 1: Mean industry-level CAPM betas estimated using 12-, 24-, and 36-month rolling windows from 1996-2023.

Figure 1 presents the evolution of mean industry-level CAPM betas estimated using rolling windows of 12, 24, and 36 months over the 1996–2023 period. Across industries and estimation horizons, mean betas are generally close to one, indicating stable exposure to aggregate market risk. Outside of crisis periods, there is little evidence of persistent upward or downward trends in average market sensitivities.

Shorter estimation windows exhibit substantially greater time-series variation in mean betas. In the 12-month window, several industries display sharp, transitory deviations, particularly during periods of market stress. These fluctuations are progressively smoothed as the lookback window increases to 24 and 36 months, consistent with longer windows filtering out high-frequency noise while preserving low-frequency movements.

A notable exception is the Agriculture, Forestry, and Fishing industry, which exhibits extreme spikes in estimated mean beta around 2020 across all horizons. These spikes are not mirrored by comparable movements in other industries and coincide with elevated dispersion and missingness, suggesting that these estimates may be influenced by sector-specific shocks or limited cross-sectional coverage rather than a structural increase in systematic risk.

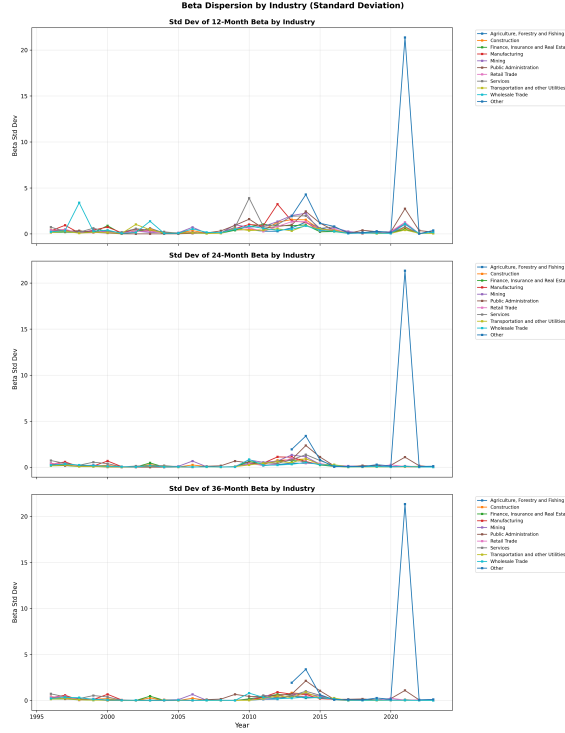


Figure 2: Cross-sectional dispersion of industry-level CAPM betas (standard deviation) using 12-, 24-, and 36-month rolling windows.

Figure 2 reports the cross-sectional standard deviation of industry betas. Dispersion is generally low during tranquil periods but rises sharply during major crises, most prominently during the 2008 financial crisis and the 2020 COVID-19 shock. This increase in dispersion is most pronounced for the 12-month window and becomes more muted as the estimation horizon lengthens, indicating that shorter windows amplify heterogeneity in beta estimates during turbulent periods.

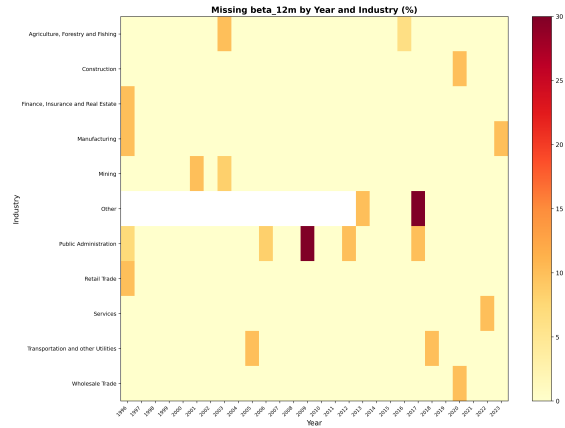


Figure 3: Percentage of missing 12-month beta estimates by year and industry.

Figure 3 shows the proportion of missing 12-month beta estimates by industry and year. Missings is generally low across most industries and time periods, but it increases episodically for specific sectors, including Agriculture, Forestry, and Fishing, Public Administration, and the residual “Other” category. These patterns suggest that extreme beta realizations and dispersion spikes for certain industries may partially reflect data availability constraints rather than underlying economic risk.

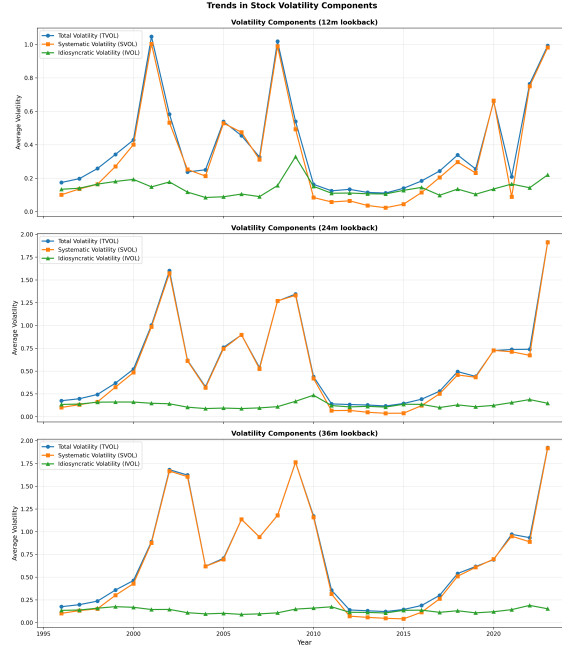


Figure 4: Average total (TVOL), systematic (SVOL), and idiosyncratic (IVOL) volatility using 12-, 24-, and 36-month rolling windows.

Figure 4 presents the decomposition of average stock volatility into total volatility (TVOL), systematic volatility (SVOL), and idiosyncratic volatility (IVOL). Across all estimation windows, TVOL and SVOL move nearly one-for-one over time, indicating that variation in total volatility is largely driven by market-wide factors. Both measures exhibit pronounced spikes during major downturns, including the early 2000s recession, the global financial crisis, and the COVID-19 shock.

In contrast, idiosyncratic volatility remains low and relatively stable throughout the sample period. While IVOL increases modestly during crisis episodes, its magnitude is small relative to systematic volatility. Longer estimation windows further smooth short-term fluctuations in all volatility components, highlighting persistent long-run volatility cycles while preserving the timing of major crisis-related peaks.

Overall, the evidence indicates that systematic risk is the dominant driver of both beta variation and aggregate volatility dynamics, while idiosyncratic risk plays a limited and comparatively stable role.